Dimensionality Reduction and Clustering

11

Pre

CS 3244
Machine Learning

NUS | Computing

Recap and Forecast

Decision Tree Learning (DTL)

- Expressive: can overfit to achieve $J_{train} = 0$
- Need pruning to combat variance

Extensions: Diversity is good

Bootstrap & Bagging: Sample with replacement to generate new $\mathcal{D}_t$.

- With each, train a h_t, apply uniform voting.
- Variance lowered through averaging.

Boosting: Make a strong classifier from simple weak classifiers.

- Reweight $\widetilde{\mathcal{D}}_t$ to account for easy / hard examples.
- Lower bias via simple hypotheses.

Random Forest: Combine *Bagging* with DTL and introduce feature projection.

- Each h_t tuned to some subset of the features.

Learning Outcomes for this week:

- Understand the basics of unsupervised learning;
- Understand different unsupervised learning algorithms for:
  - Dimensionality reduction: For aiding visualization, exploratory analysis.
  - Clustering: Finding latent patterns.
- Implement and understand k means clustering;
- Understand the Expectation Maximization (EM) meta-algorithm;
- Apply unsupervised learning on real-world data.

Dimensionality Reduction

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Big Data

Input data may have thousands or millions of dimensions!

- e.g., facial images, natural language text, stock market time series

Can we describe them more succinctly? That would help…

- easier learning – fewer parameters
- visualization – hard to visualize more than 3D or 4D
- discover "intrinsic dimensionality" of data
  - high dimensional data that is truly lower dimensional

Dimensionality reduction

Example: Stock market data

Presence of inherent structure:

- Certain equities might change in a similar way
  - E.g., Transshipment stocks, Regional stocks
- Use less information to get a good approximation

Photo credit: Andreas Poike @ Flickr

Lower dimensional projections

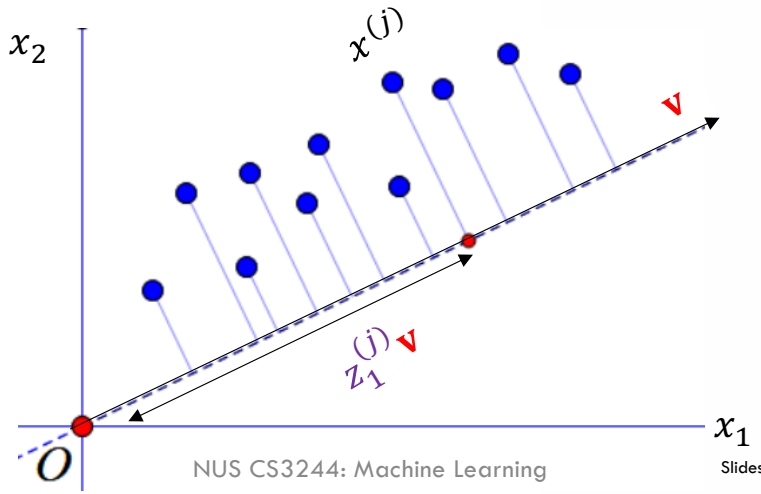
Rather than picking a subset of the features, we can construct new features that are combinations of existing features. Let's see this in the unsupervised setting (just $\mathbf{X}$, drop y).

Example: $\mathbf{x}^{(j)}: \{x_1, x_2\} \rightarrow \Phi_{\mathbf{v}}(\mathbf{x}): \{z_1^{(j)}\}$

Same for all points

Linear function $\Phi_{\mathbf{v}}: \{x_1, x_2\} = z_1 \cdot \mathbf{v} = z_1 \cdot \{v_1, v_2\}$

Closest point on $\mathbf{v}$ to original $\mathbf{x}$



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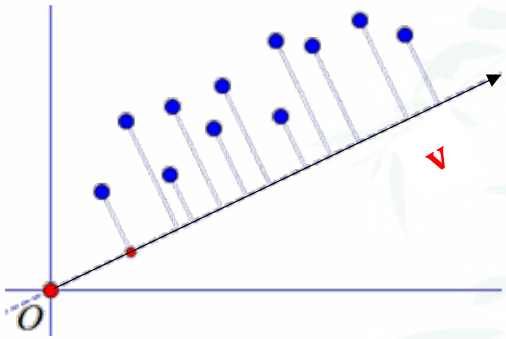
Let's Analyze a 2D example

Original 2D points: $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$

Project on $\mathbf{v}$, get 1D points:

$$\{z_1^{(1)}, \dots, z_1^{(n)}\}$$

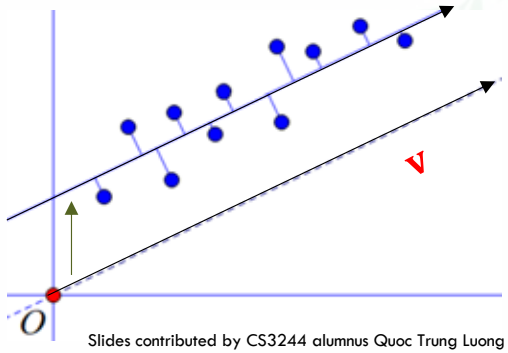
* *The reference point could be any point on* $\mathbf{v}$



By projection, we can lose some information.

Reconstruct the original data by shifting $\mathbf{v}$ so that the sum of squares of the distance from the original data point is minimized.

Direction that leads to smallest error is most desirable.



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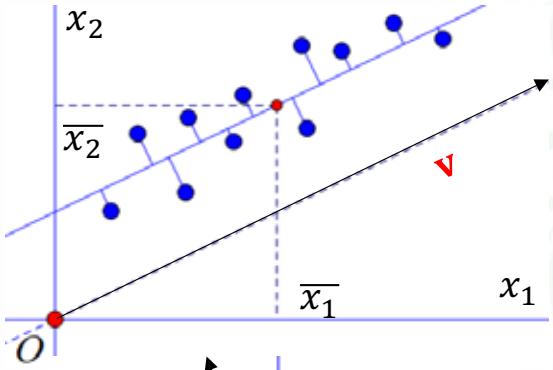
Observation

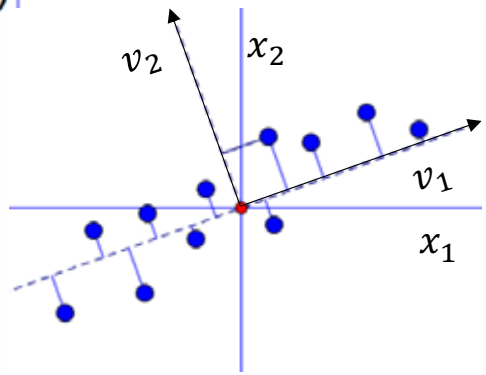
The closest line passes through center of the points

- Conveniently reconstruct original points by demean them, project to $\mathbf{v}$, adding $\bar{\mathbf{x}}$ back
- Reconstruction point: $\mathbf{x}_{rec}^{(j)} = \bar{\mathbf{x}} + z^{(j)} \cdot \mathbf{v}$



The procedure can be viewed as:

- Transform to new coordinate $\{x_1, x_2\} \rightarrow \{v_1, v_2\}$ (The new origin is at the center of the points)
- Drop unimportant dimensions $\{v_1, v_2\} \rightarrow \{v_1\}$
- Reconstruct by:
  $\{v_1\} \rightarrow \{v_1, 0\} \xrightarrow{Inverse\ Transform} \{x_1, x_2\}$



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Linear projection

Project a point into another space:

Row vectors

1. Point: $\mathbf{x} = \{x_1, \dots, x_n\}$
2. Select a basis: a set of basis vectors: $\{\mathbf{v}_1, \dots, \mathbf{v}_N\}$
3. Let's consider only orthonormal bases: e.g., $\mathbf{v}_i \mathbf{v}_i^\top = 1$ and $\mathbf{v}_i \mathbf{v}_j^\top = 0$ for $i \neq j$
4. Select a center: $\bar{\mathbf{x}}$ defines the offset of the space
5. Represent $\mathbf{x}$ in new basis: $\{z_1, \dots, z_N\}$ where $z_k = (\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_k^\top$
6. Projected point: $\{z_1, \dots, z_K\}$

Principal Component Analysis

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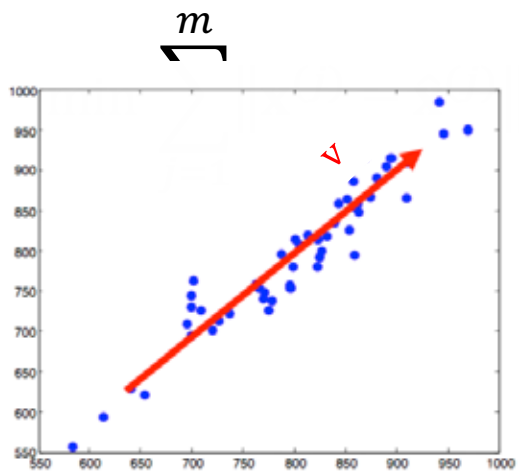
Principal component analysis

Project n-dimensional data into K-dimensional space while preserving information:

- e.g., project space of 30,000 words into 50-dimensions
- e.g., project 2D into 1D

Compute projection $\mathbf{v}$ with minimum least squares reconstruction error:



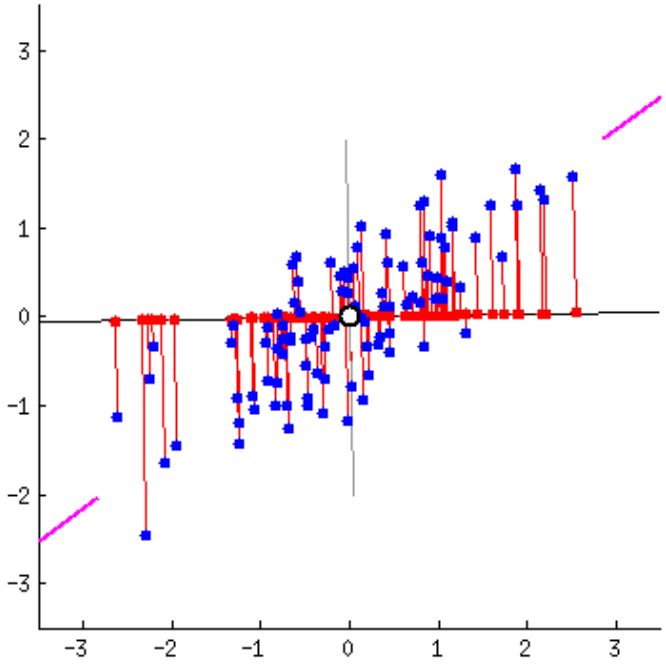
$$\overbrace{\quad}^{m} \; ? = \min \sum_{j=1}^{m} \left\| \mathbf{x}^{(j)} - (\bar{\mathbf{x}} + \sum_{k=1}^{K} z_k^{(j)} \mathbf{v}_k) \right\|^2$$

Choose $\mathbf{v}$ to minimize residual variance

$\mathbf{v}$ is thus also the direction of maximum variance

Diagrams modified from Alex Ihler's slides @ UCI

Choosing representative axes



Reconstruction Cost J_{recon}

A point is exactly represented by:

$$\mathbf{x}^{(j)} = \bar{\mathbf{x}} + \sum_{k=1}^{n} z_k^{(j)} \mathbf{v}_k$$

But in PCA, we only reconstruct with $K < n$ dimensions:

$$\hat{\mathbf{x}}^{(j)} = \bar{\mathbf{x}} + \sum_{k=1}^{K} z_k^{(j)} \mathbf{v}_k$$

The cost then is: $J_{recon} = \sum_{j=1}^{m} \left\| \mathbf{x}^{(j)} - \hat{\mathbf{x}}^{(j)} \right\|^2$

$$\underset{(m \times n)}{\bar{\mathbf{X}}} = \begin{bmatrix} \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \end{bmatrix}$$

Reconstruction Cost J_{recon}

A point is exactly represented by: $\mathbf{x}^{(j)} = \bar{\mathbf{x}} + \sum_{k=1}^{n} z_k^{(j)} \mathbf{v}_k$

But in PCA, we only reconstruct with $K < n$ dimensions: $\hat{\mathbf{x}}^{(j)} = \bar{\mathbf{x}} + \sum_{k=1}^{K} z_k^{(j)} \mathbf{v}_k$

The cost then is: $J_{recon} = \sum_{j=1}^{m} \left\| \mathbf{x}^{(j)} - \hat{\mathbf{x}}^{(j)} \right\|^2$

$$\bar{\mathbf{X}} = \underset{(m \times n)}{} \begin{bmatrix} \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \end{bmatrix}$$

$$= \sum_{j=1}^{m} \left\| \left(\bar{\mathbf{x}} + \sum_{k=1}^{n} z_k^{(j)} \mathbf{v}_k \right) - \left(\bar{\mathbf{x}} + \sum_{k=1}^{K} z_k^{(j)} \mathbf{v}_k \right) \right\|^2$$

$$= \sum_{j=1}^{m} \left\| \left(\sum_{k=K+1}^{n} z_k^{(j)} \mathbf{v}_k \right) \right\|^2 = \sum_{j=1}^{m} \left[\left(\sum_{k=K+1}^{n} z_k^{(j)} \underbrace{\mathbf{v}_k \mathbf{v}_k^\top}_{1} z_k^{(j)} \right) + \sum_{\substack{K+1 \le h,\, l \le n \\ h \ne l}} z_h^{(j)} \underbrace{\mathbf{v}_h \mathbf{v}_l^\top}_{0} z_l^{(j)} \right]$$

Reconstruction Cost J_{recon}

$$J_{recon} = \sum_{j=1}^{m}\left[\left(\sum_{k=K+1}^{n} z_k^{(j)} \underbrace{\mathbf{v}_k\mathbf{v}_k^\top}_{1} z_k^{(j)}\right) + \sum_{\substack{K+1\le h,\, l\le n\\ h\neq l}} z_h^{(j)} \underbrace{\mathbf{v}_h\mathbf{v}_l^\top}_{0} z_l^{(j)}\right]$$

$$= \sum_{j=1}^{m}\sum_{k=K+1}^{n} {z_k^{(j)}}^2 = \sum_{k=K+1}^{n}\sum_{j=1}^{m} {z_k^{(j)}}^2$$

$$= \sum_{k=K+1}^{n}\sum_{j=1}^{m} \mathbf{v}_k(\mathbf{x}-\bar{\mathbf{x}})^\top(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_k^\top$$

$$= \sum_{k=K+1}^{n} \mathbf{v}_k(\mathbf{X}-\bar{\mathbf{X}})^\top(\mathbf{X}-\bar{\mathbf{X}})\,\mathbf{v}_k^\top = m\sum_{k=K+1}^{n} \mathbf{v}_k\Sigma\mathbf{v}_k^\top$$

$$\underset{(m\times n)}{\bar{\mathbf{X}}} = \begin{bmatrix} \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \\ \dots & \bar{\mathbf{x}} & \dots \end{bmatrix}$$

Covariance matrix

This choice of v_k leads to a minimum J_{recon}

Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be eigenvectors of Σ and the corresponding eigenvalues be $\{\lambda_1, \dots, \lambda_n\}$.

$$J_{recon} = m\sum_{k=K+1}^{n} \mathbf{v}_k\,\lambda_k\mathbf{v}_k^\top = m\sum_{k=K+1}^{n}\lambda_k$$

Better choose $\{\mathbf{v}_1, \dots, \mathbf{v}_K\}$ corresponding to highest eigenvalues

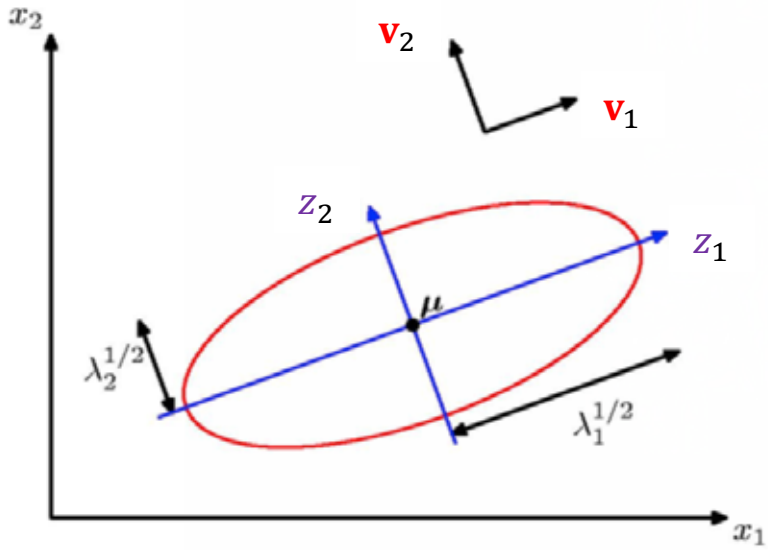
Back to the Gaussian…

$$\Delta^2 = (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu)$$

(Eigen)decompose covariance matrix $\Sigma = \mathrm{V}\Lambda\mathrm{V}^\top$

Write Σ in terms of eigenvectors:

$$\Sigma^{-1} = \Sigma_{j=1}^{n} \frac{1}{\lambda_j} \mathbf{v}_j \mathbf{v}_j^\top$$



Diagrams modified from Alex Ihler's slides @ UCI

Basic PCA algorithm

1. Start from $m \times n$ data matrix $\mathbf{X}$
2. Center: subtract μ from each row of $\mathbf{X}_c \leftarrow \mathbf{X} - \overline{\mathbf{X}}$
3. (Typically) scale each dimension by its variance
   To decorrelate the magnitude of the variable
4. Compute covariance matrix: $\Sigma \leftarrow \frac{1}{m}\mathbf{X}_c^\top \mathbf{X}_c$
5. Find eigenvectors and eigenvalues of Σ
6. Compute K eigenvectors with highest eigenvalues

$$\Sigma = \mathbf{V}\Lambda\mathbf{V}^\top$$

Problem: can be slow when $\mathbf{X}$ is big

Singular Value Decomposition

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Singular Value Decomposition

One method to calculate the orthogonal decomposition of $\mathbf{X}$ in PCA.

Nice as we can obtain the K largest directly.

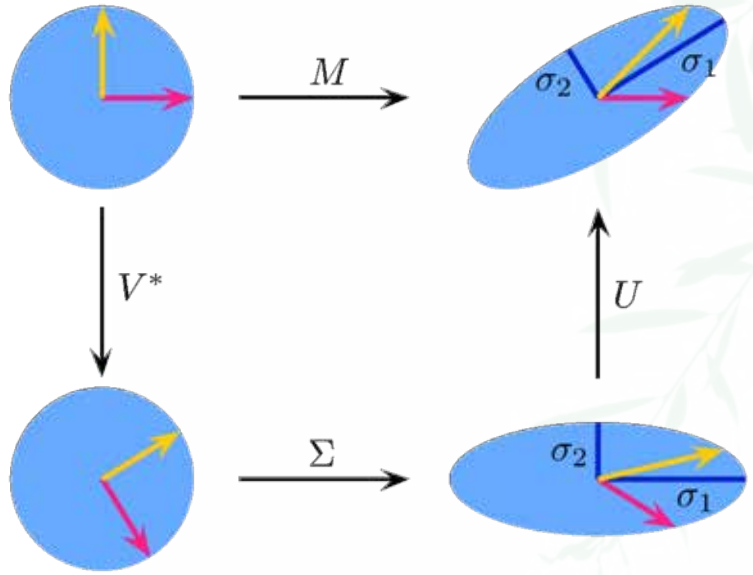
Nice because <u>all</u> matrices work with SVD.

- Other factorizations don't work with certain matrices (e.g., non-square).

Geometry of SVD

"A singularly valuable decomposition" – Dan Kalman

*Any transformation (matrix) can be thought of as a rotation V, followed by scaling Σ, followed by rotation U.**



$$M = U \cdot \Sigma \cdot V$$

* *For square matrices only; the rotation interpretation doesn't work for rectangular matrices.*

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Singular Value Decomposition

Write $\mathbf{X} = U \Sigma V^\top$.

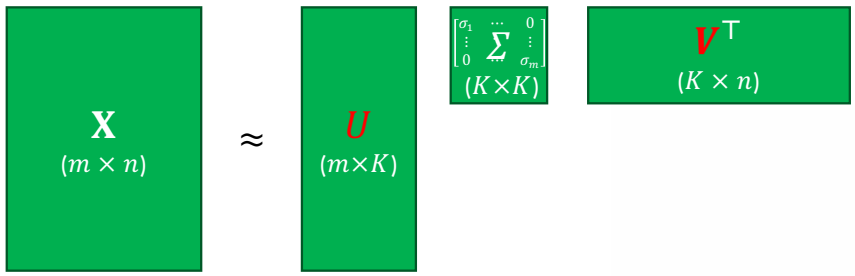
U: weight matrix, one row per point – coordinate of $\mathbf{x}^{(j)}$ in eigenspace

Σ: singular value matrix, a diagonal matrix

- each entry is an eigenvalue λ_k

$V^\top$: singular vector matrix

- each row is an eigenvector $\mathbf{v}_k$



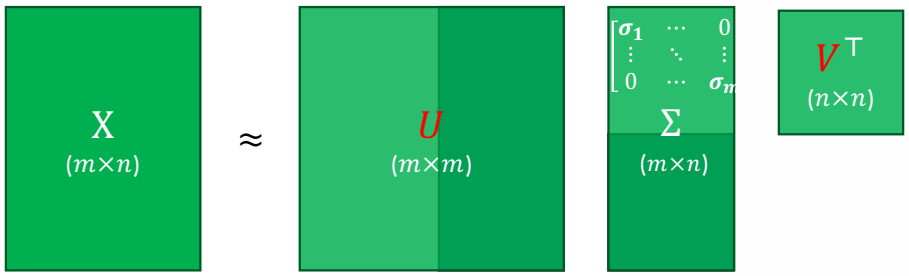
Singular Value Decomposition

Write $\mathbf{X} = U\,\Sigma\,V^\top$.

- U: weight matrix, one row per point – coordinate of $\mathbf{x}^{(j)}$ in eigenspace
- Σ: singular value matrix, a diagonal matrix
  - each entry is an eigenvalue λ_k
- $V^\top$: singular vector matrix
  - each row is an eigenvector $\mathbf{v}_k$

Orthogonal:

$$\mathbf{X}^\top\mathbf{X} = V\,\Sigma\,\Sigma\,V^\top = V\,D\,V^\top$$
$$\mathbf{X}\mathbf{X}^\top = U\,\Sigma\,\Sigma\,U^\top = U\,D\,U^\top$$



PCA using SVD algorithm

1. Start from $m \times n$ data matrix $\mathbf{X}$
2. Re-center: subtract μ from each row of $\mathbf{X}_c \leftarrow \mathbf{X} - \overline{\mathbf{X}}$
3. (Typically) scale each dimension by its variance
   To decorrelate the magnitude of variable
4. Call SVD algorithm on $\mathbf{X}_c$ – ask for K singular vectors

Principal components: K singular vectors with highest singular values (rows of $V^\top$)

Coefficients are derived from $U\Sigma$ matrix:

$$\mathbf{x}^{(j)} = U_{j,1}\Sigma_{11}\mathbf{v}_1 + U_{j,2}\Sigma_{22}\mathbf{v}_2 + \cdots + U_{j,K}\Sigma_{KK}\mathbf{v}_K$$

PCA: Eigenfaces

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Eigenfaces

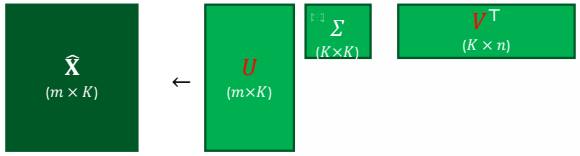
Eigen-$\mathbf{X}$ paradigm = represent $\mathbf{X}$ using PCA

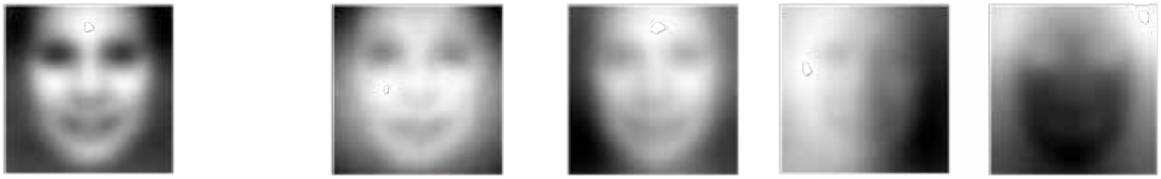
"What about faces?" (Turk and Pentland, 91)

Decomposition:

μ v_1 v_2 v_3 v_4 ...

Eigenfaces: Reconstruction





μ $\quad v_1 \quad v_2 \quad v_3 \quad v_4 \quad \cdots$

$K = 4$

$K = 50$

$K = 4$

$K = 50$

SVD and PCA relationship

When the data is centered (mean 0) then PCA via SVD on all n dimensions yields the same result.

We typically want dimensionality reduction, so we choose $K \ll n$.

So no need to compute the full covariance matrix.

We also rarely have a square data matrix $\mathbf{X}$.

$\therefore$ Use SVD all the time!

K Means Clustering

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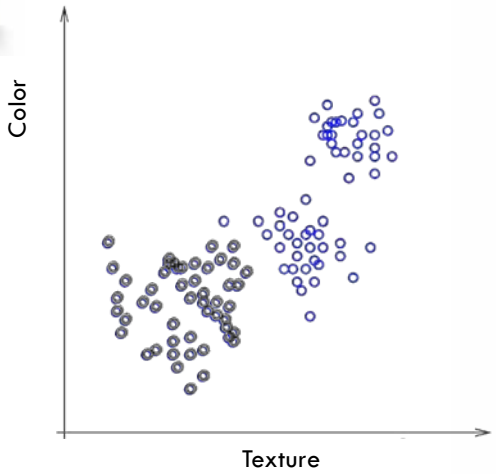
Clustering



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Clustering

A cluster is a collection of points S.

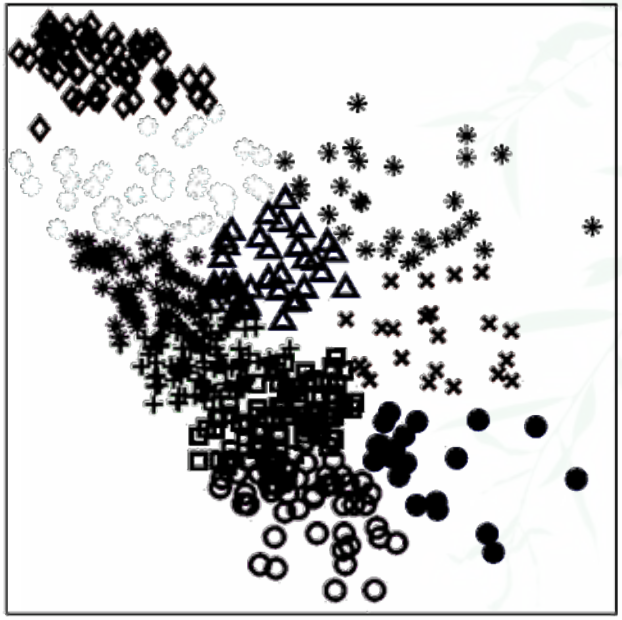
A K-clustering is a partition of the data into K clusters $S_1, S_2, \ldots, S_K$.

$$\bigcup_{k=1}^{K} S_k = \mathcal{D}$$

Hard clustering assigns each point into exactly one cluster:

$$S_i \cap S_j = \emptyset, \text{ for } i \neq j.$$

Each cluster has a center μ_k.



Clustering Quality

Points in a cluster should be similar (close to each other, and to the center)

Defines the error measure as well.

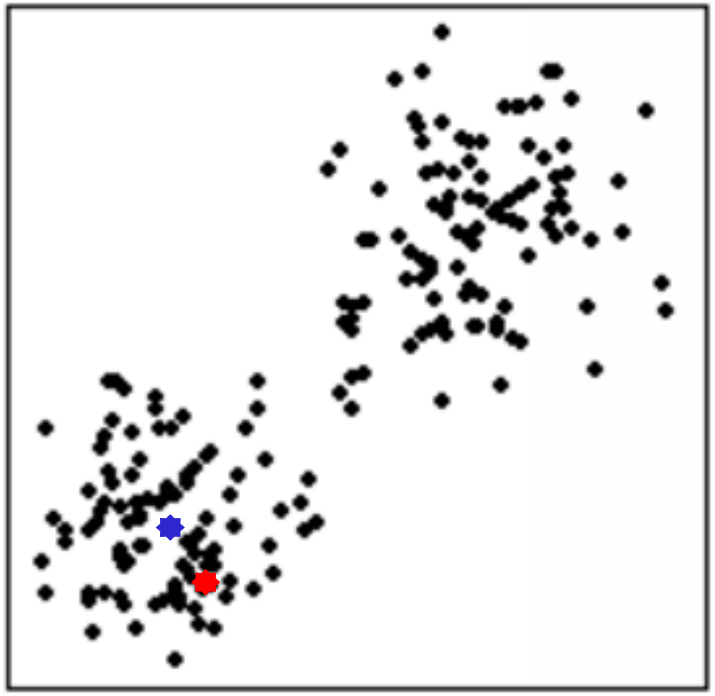
Error in cluster k: $J_k = \sum_{\mathbf{x}^{(j)} \in S_k} \left|\left|\mathbf{x}^{(j)} - \mu_k\right|\right|^2$.

Other measures (e.g., L^p norms) can be used as well

Minimizing $J_{clust} = \sum_{k=1}^{K} J_k$ is not convex, known to be NP hard.

But simple, good solutions exist.

K means, Illustrated



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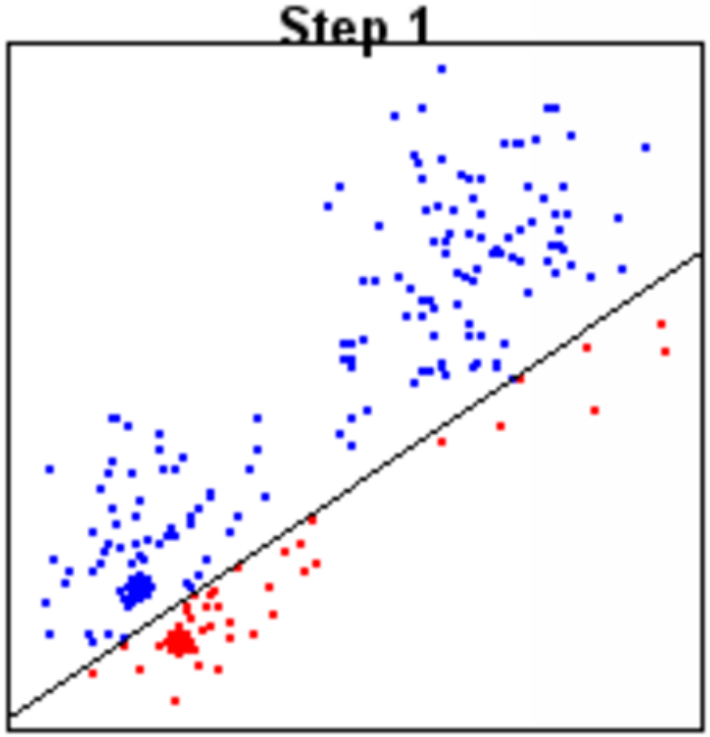
Assign points to clusters



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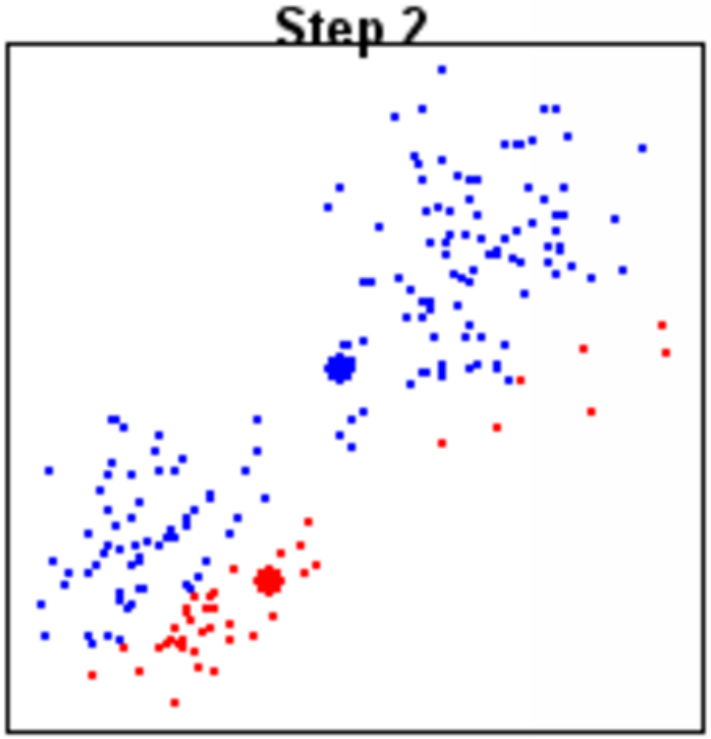
Recalculate cluster centers



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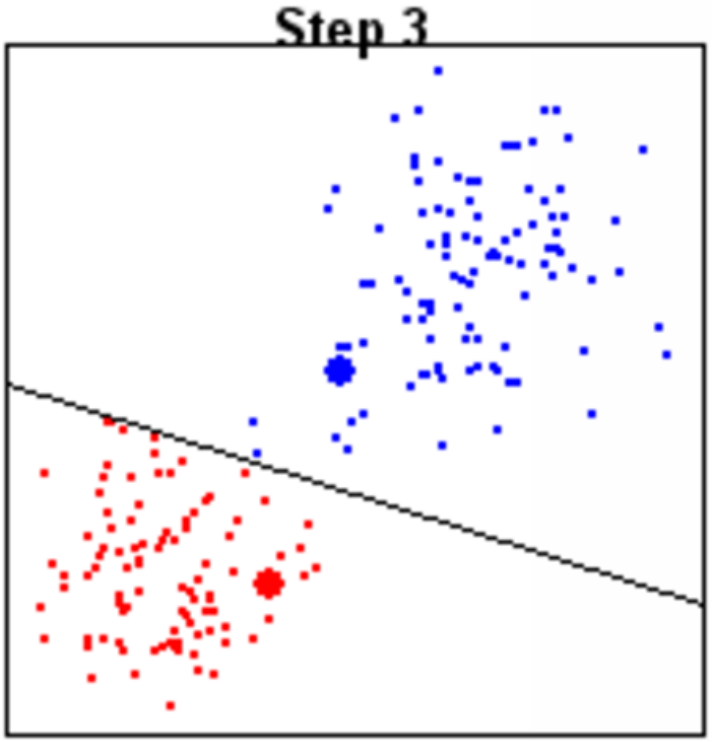
Reassign points to clusters



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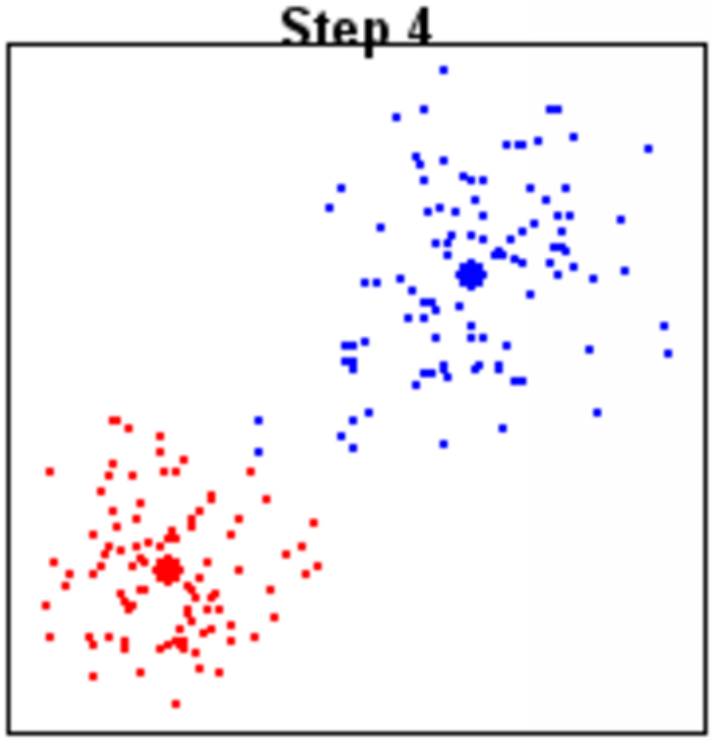
Recalculate cluster centers



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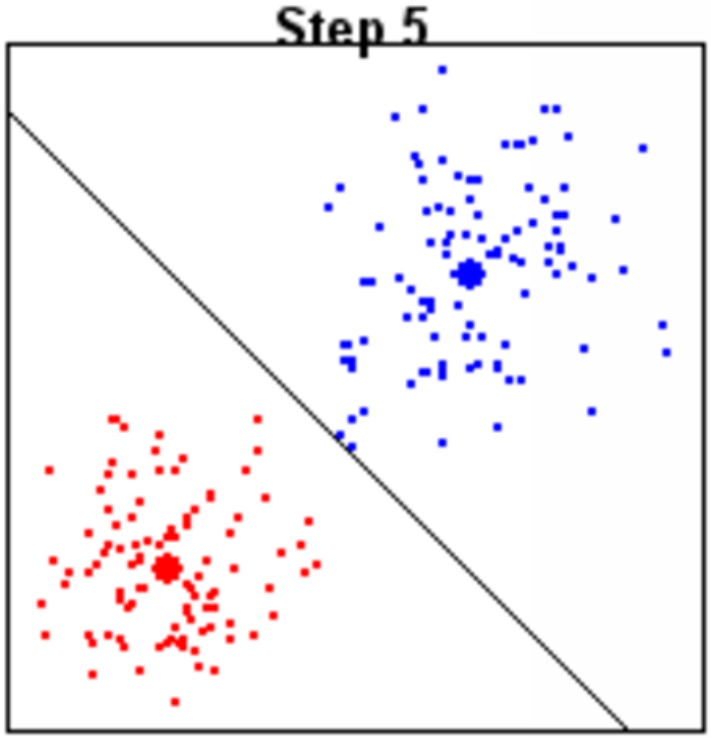
Reassign points to clusters



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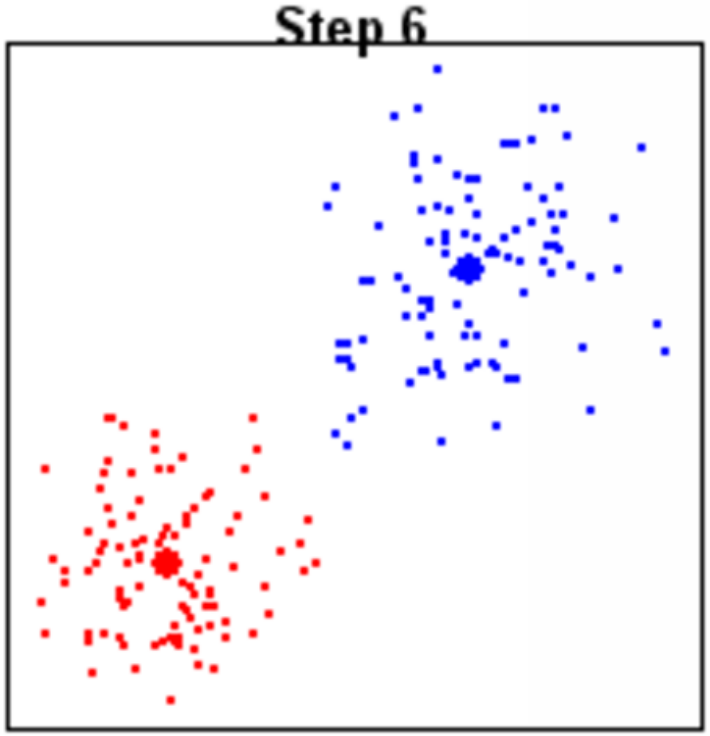
…Until convergence



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K means

for $k = 1$ to K do: # initialize randomly

  $\mu_k \leftarrow$ some random location

repeat until converged:

  for $j = 1$ to m do:

    $y^{(j)} \leftarrow \arg\min_k \left\|\mu_k - \mathbf{x}^{(j)}\right\|$ # A. assign example

  for $k = 1$ to K do:

    $\mu_k \leftarrow \frac{1}{m_k} \sum_{\mathbf{x}^{(j)} \in k} \mathbf{x}^{(j)}$ # B. re-estimate center

return $\boldsymbol{y}$

Proof of convergence

Only lines A and B change values of μ and y. We can show that both can only decrease J_{clust}.

J_{clust} starts with a value after initialization and lower bounded by 0.

As y is discrete, there can only be a finite number of steps if each step can only decrease J_{clust}.

Line A: $\left\|\mathbf{x}^{(j)} - \mu(t)\right\| \leq \left\|\mathbf{x}^{(j)} - \mu(t-1)\right\|$

Line B: μ_k calculated by mean, which minimizes MSE.

Decreasing J_{clust} (details, not in video)

Line A: Assign each point to its closest cluster

$\rightarrow$ unchanged/decreased J_{clust}

Line B: For each k, choose μ_k to minimize: $J_k = \sum_{x^{(j)} \in S_k} \left\| \mathbf{x}^{(j)} - \mu_k \right\|^2$

Let center of points in cluster k be $\bar{x}_k = \frac{1}{m_k} \sum_{x^{(j)} \in S_k} x^{(j)}$

$$J_k = \sum_{\mathbf{x}^{(j)} \in S_k} (\mathbf{x}^{(j)} - \bar{\mathbf{x}}_k + \bar{\mathbf{x}}_k - \mu_k)^2$$

$$= \sum_{\mathbf{x}^{(j)} \in S_k} (\mathbf{x}^{(j)} - \bar{\mathbf{x}}_k)^2 + \sum_{x^{(j)} \in S_k} (\bar{\mathbf{x}}_k - \mu_k)^2 \geq \sum_{\mathbf{x}^{(j)} \in S_k} (\mathbf{x}^{(j)} - \bar{\mathbf{x}})^2$$

Setting $\mu_k = \bar{\mathbf{x}}$ $\rightarrow$ unchanged/decreased J_{clust}

We'll discuss this more in tutorial.

K Means Extensions

CS3244 Machine Learning

NUS National University of Singapore
Department of Computer Science
School of Computing

Initialization

… is an issue, but quite surmountable.

Furthest-first heuristic: after picking first point at random, pick all remaining $k-1$ centers to be as far as possible from all previously selected centers, i.e.:

$$m = \arg\max_j \min_{k'<k} \left\|\mathbf{x}^{(j)} - \mu_{k'}\right\|^2$$

Then set $\mu_k = \mathbf{x}^{(j)}$.

Deterministic after the first point selection.

Adding randomness

k-means++*: When selecting μ_k, choose point at random, but in probability proportional to squared distance.

for $k = 2$ to K do:

$d^{(j)} \leftarrow \min_{k'<k} \left|\left|\mathbf{x}^{(j)} - \mu_{k'}\right|\right|^2, \forall j$ # compute distances

$\mathbf{P}: p(j) \leftarrow \frac{d^{(j)}}{\sum_{j=1}^{m} d^{(j)}}, j \in \{1, \dots, m\}$ # make prob. distribution

$c \sim \mathbf{P}$ # pick an example as the center

$\mu_k \leftarrow \mathbf{x}^{(c)}$

Run k-means using $\mu_{1..K}$ as the initial centers.

*(Standard initialization in sklearn)

Choosing K

"Should I add another cluster?"

<u>Quick Question</u>: Using J_{clust} as a fitness measure is no good alone because it will always prefer:

1) Only 1 cluster for all points ($K = 1$).

2) The same number of clusters that I'm using now, K.

3) m clusters, one for each point.

Regularizing K means

Augment another term for optimization to ensure value of k is not overly complex. With:

Bayes Information Criterion (BIC): $\arg\min_{K} J_{clust}^{(K)} + K \log n$

Akaike Information Criterion (AIC): $\arg\min_{K} J_{clust}^{(K)} + 2Kn$

Stronger regularizer, especially for high dimensional data

…where n is the dimensionality of $\mathbf{x}$.